

Lab Assignment

1. Write a program to display "hello world" in C.

Answer:

```
#include<stdio.h>

int main()
{
    printf("Hello World")
    return 0;
}
```

Output:

Hello, World!

2. Write a C program that takes two integers as input and prints their sum.

Answer:

```
#include<stdio.h>

int main()
{
    int num1 = 7, num2 = 9, sum ;
    sum = num1 + num2 ;
    printf("The value :%d",sum);
    return 0;
}
```

Output:

Enter two integers:16

3. Write a program to calculate the area of triangle [eq: $\text{area} = 0.5 * \text{height} * \text{base}$].

Answer:

```
#include <stdio.h>
```

```
int main() {
```

```
    float base=7, height=12, area;
```

```
    area = 0.5 * height * base ;
```

```
    printf("Area of the triangle:%f%f\n",area);
```

```
    return 0;
```

```
}
```

Output:

Area of the triangle:42.00

4. Write a program to calculate area of a circle having its radius (r=5).

Answer:

```
#include<stdio.h>
```

```
int main() {
```

```
    float radius = 5, area;
```

```
    area = 3.14 * radius * radius;
```

```
    printf("The area of the circle is: %.2f\n", area);
```

```
    return 0;
```

```
}
```

Output:

The area of the circle is: 78.54

5. Write a program to calculate area of an ellipse having its axes (minor 4cm, major=6cm).

Answer:

```
#include <stdio.h>
```

```
int main() {
```

```
    float minor = 4, major = 6, area;
```

```
    area = 3.14 * minor * major;
```

```
    printf("The area of the ellipse is: %.2f\n", area);
```

```
    return 0;
```

```
}
```

Output:

The area of the ellipse is: 75.36

6. Write a program to calculate simple interest for a given $P=4000$, $T=2$, $R=5.5$. ($I = P \cdot T \cdot R / 100$)

Answer:

```
#include <stdio.h>
```

```
int main() {
```

```
    float P = 4000, T = 2, R = 5.5, interest;
```

```
    interest = (P * T * R) / 100;
```

```
    printf("The simple interest is: %.2f\n", interest);
```

```
    return 0;
```

```
}
```

Output:

The simple interest is: 440.00

7. Write a C program that takes a temperature in Celsius as input and converts it to Fahrenheit. The formula is: $F = (C * 9/5) + 32$

Answer:

```
#include <stdio.h>
```

```
int main() {
```

```
    float celsius, fahrenheit;
```

```
    printf("Enter temperature in Celsius: ");
```

```
    scanf("%f", &celsius);
```

```
    fahrenheit = (celsius * 9/5) + 32;
```

```
    printf("Temperature in Fahrenheit: %.2f\n",  
fahrenheit);
```

```
    return 0;
```

```
}
```

Output: Enter temperature in Celsius: 26

Temperature in Fahrenheit: 78.80